

Drawing drills — line codes, scrambling, decoding, waves

Untimed practice on paper. The answer key starts on a new page.

Part A — Draw the line codes

Fill every cell of each grid. Each problem states the slide conventions to use.

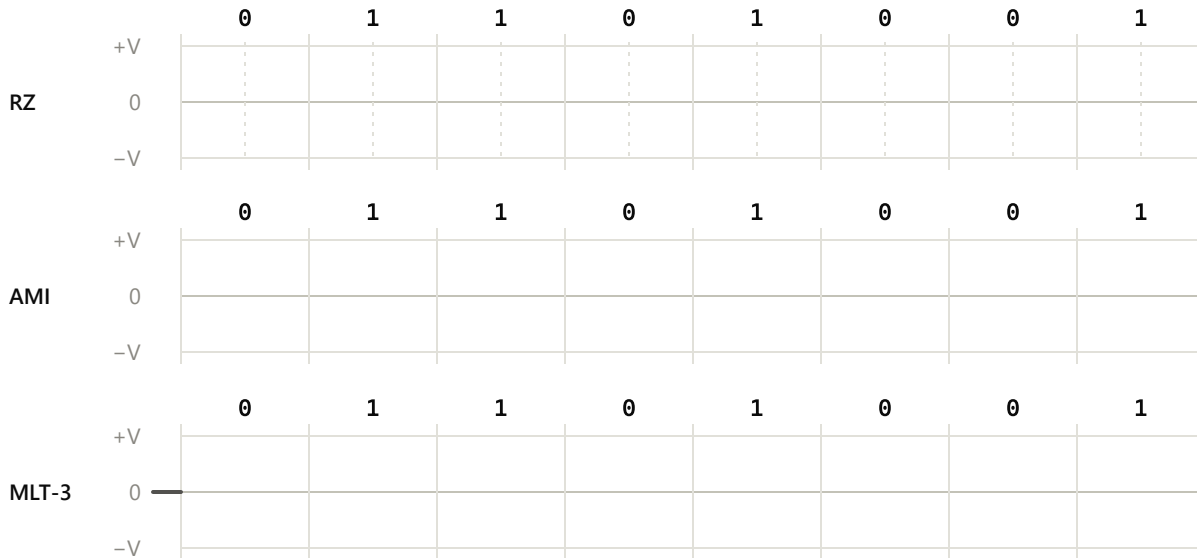
1. Draw the bit stream **10011010** as a waveform in each of the line codes below (fill every cell of the grid). Use the slide conventions:

- NRZ-L (slide figure): 0 = +V, 1 = -V.
- NRZ-I: a 1 inverts the level at the start of the bit, a 0 does not; assume the level before the first bit is +V.
- Manchester: 0 = high→low, 1 = low→high (a transition in the middle of every bit).
- Differential Manchester: always a transition in the middle of the bit; a 0 also has a transition at the start of the bit, a 1 does not; assume the level before the first bit is +V.

	1	0	0	1	1	0	1	0
NRZ-L	+V							
	0							
	-V							
NRZ-I	+V							
	0							
	-V							
Manchester	+V							
	0							
	-V							
Differential Manchester	+V							
	0							
	-V							

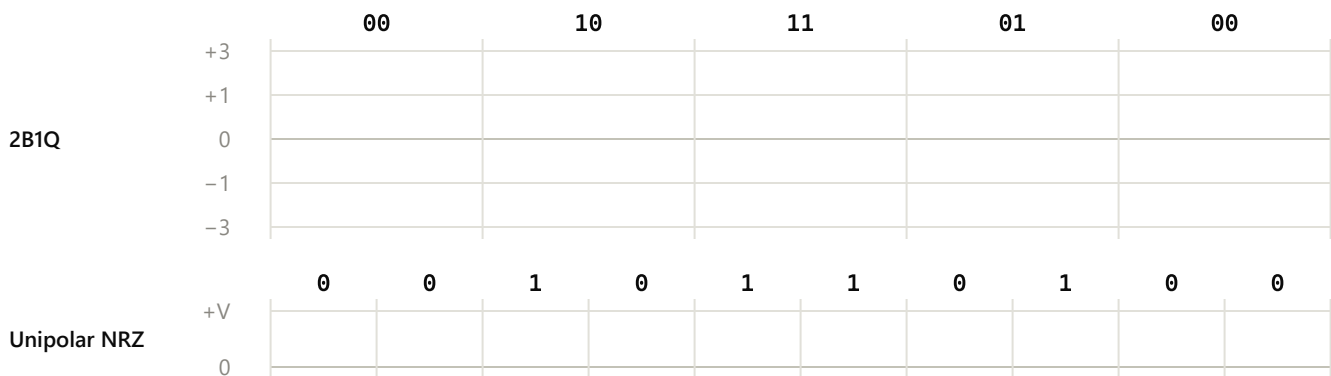
2. Draw the bit stream **01101001** as a waveform in each of the line codes below (fill every cell of the grid). Use the slide conventions:

- RZ: a 1 is +V then 0 V, a 0 is -V then 0 V (every bit returns to zero in the middle).
- AMI: 0 = 0 V; the 1s alternate +V, -V, ..., starting with +V.
- MLT-3: a 0 keeps the level; a 1 moves round the cycle 0 → +V → 0 → -V → ...; start at 0 V with the last nonzero level -V, so the first 1 goes to +V.



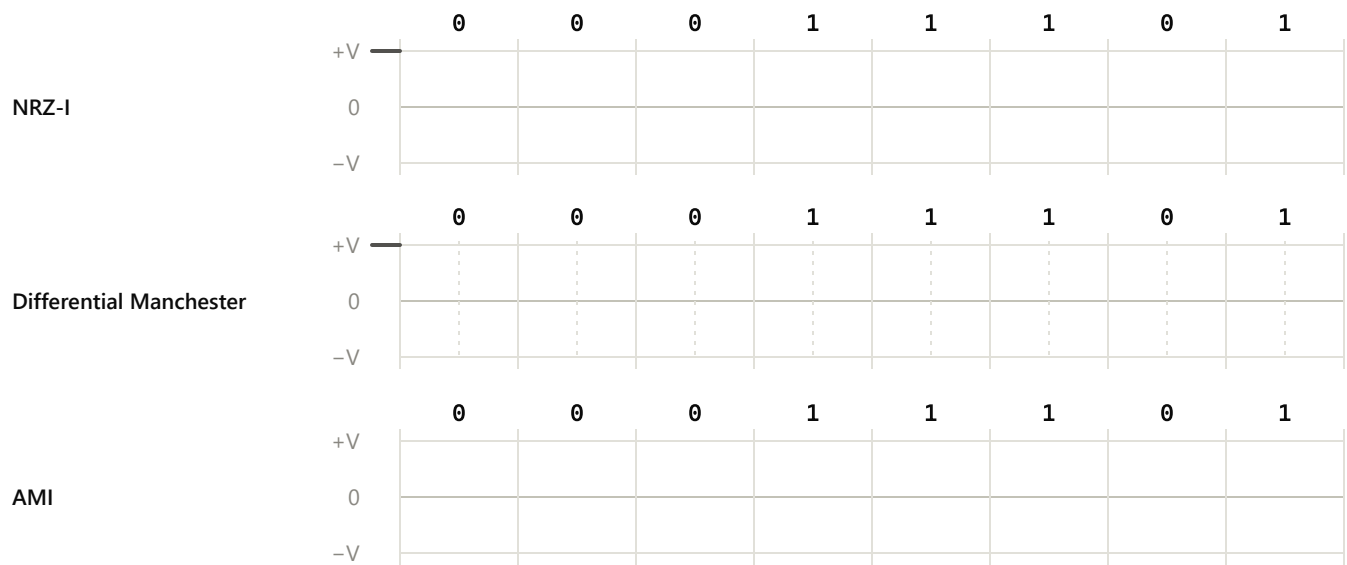
3. Draw the bit stream **0010110100** as a waveform in each of the line codes below (fill every cell of the grid). Use the slide conventions:

- 2B1Q: one level per pair of bits from the transition table (previous level positive: 00 → +1, 01 → +3, 10 → -1, 11 → -3; previous level negative: the signs reversed); assume the level before the first pair is positive.
- Unipolar NRZ: 1 = +V, 0 = 0 V.



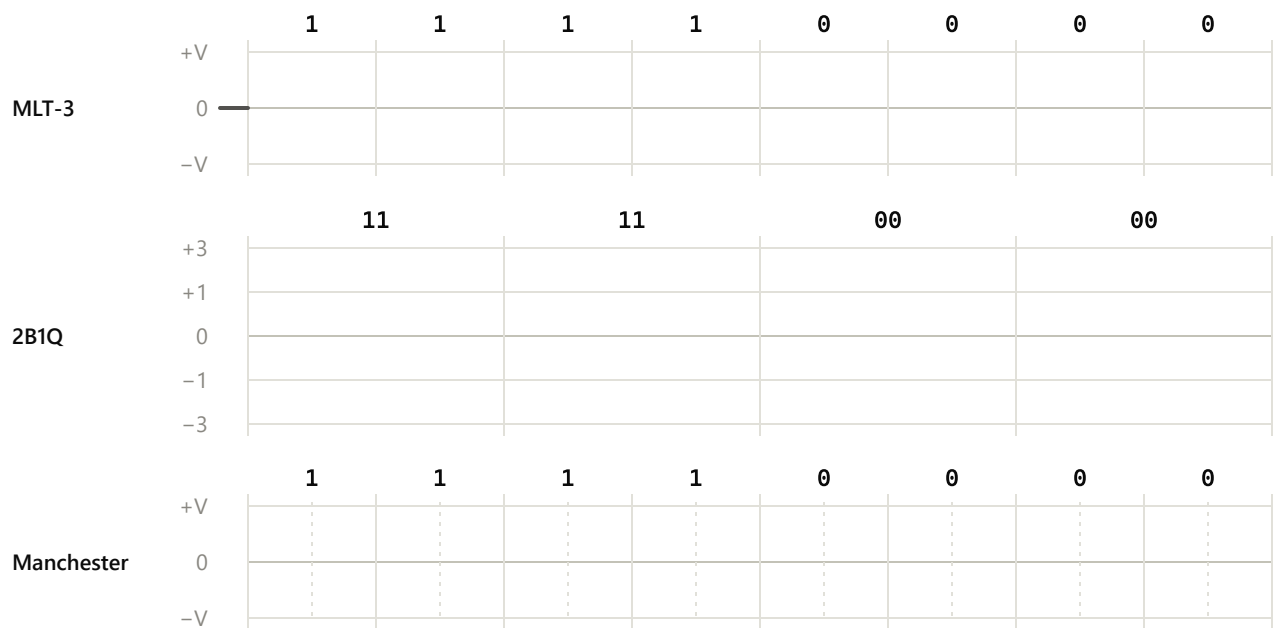
4. Draw the bit stream **00011101** as a waveform in each of the line codes below (fill every cell of the grid). Use the slide conventions:

- NRZ-I: a 1 inverts the level at the start of the bit, a 0 does not; assume the level before the first bit is +V.
- Differential Manchester: always a transition in the middle of the bit; a 0 also has a transition at the start of the bit, a 1 does not; assume the level before the first bit is +V.
- AMI: 0 = 0 V; the 1s alternate +V, -V, ..., starting with +V.



5. Draw the bit stream **11110000** as a waveform in each of the line codes below (fill every cell of the grid). Use the slide conventions:

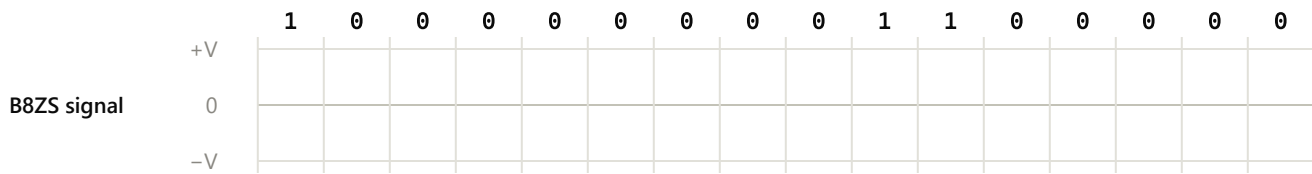
- MLT-3: a 0 keeps the level; a 1 moves round the cycle 0 → +V → 0 → -V → ...; start at 0 V with the last nonzero level -V, so the first 1 goes to +V.
- 2B1Q: one level per pair of bits from the transition table (previous level positive: 00 → +1, 01 → +3, 10 → -1, 11 → -3; previous level negative: the signs reversed); assume the level before the first pair is positive.
- Manchester: 0 = high→low, 1 = low→high (a transition in the middle of every bit).



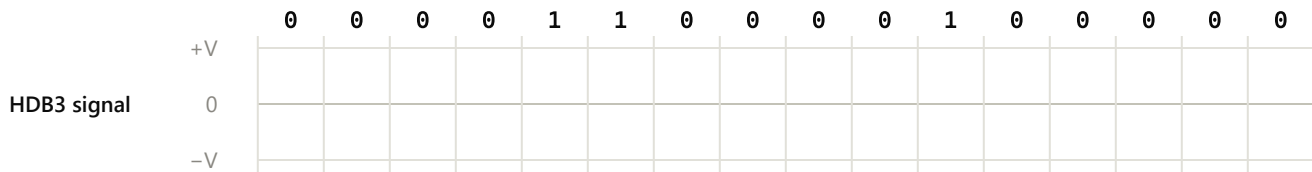
Part B — Scrambling (B8ZS and HDB3)

Write the AMI pulses first, then substitute. Mark each V and B.

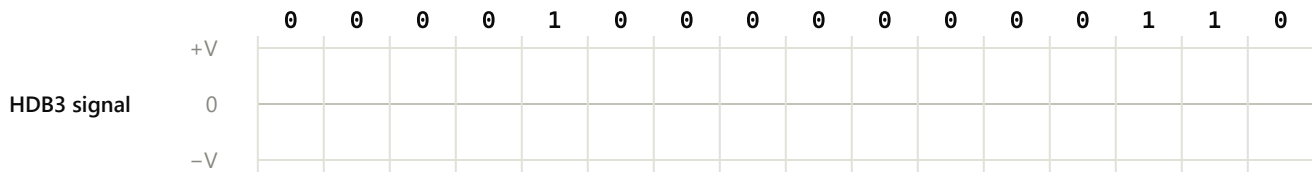
6. Apply **B8ZS** scrambling (on top of AMI) to the data stream **1000 0000 0110 0000** and draw the line signal. The last nonzero pulse sent before this stream was negative ($-V$).



7. Apply **HDB3** scrambling (on top of AMI) to the data stream **0000 1100 0010 0000** and draw the line signal. The last nonzero pulse sent before this stream was positive ($+V$). At the start of the stream the number of nonzero pulses since the last substitution is even.



8. Apply **HDB3** scrambling (on top of AMI) to the data stream **0000 1000 0000 0110** and draw the line signal. The last nonzero pulse sent before this stream was negative ($-V$). At the start of the stream the number of nonzero pulses since the last substitution is odd.



Part C — Decode the signal

Write the bit string that was sent.

9. A receiver sees this **Manchester** signal (two half-bit levels per bit, grouped per bit, left to right): $(+-)(-+)(-+)(-+)(+-)(+-)(-+)(+-)$. The direction of the mid-bit transition gives the bit: high→low = 0, low→high = 1. What bit string was sent?

WORK

Bits sent = _____

10. A receiver sees this **NRZ-I** signal (one level per bit, left to right): $- - + + + - + -$. The level before the first bit was $+V$; a change of level at the start of a bit means 1, no change means 0. What bit string was sent?

WORK

Bits sent = _____

11. A receiver sees this **MLT-3** signal (one level per bit, left to right): $+ 0 0 - - 0 + +$. The signal started at $0 V$. A change of level means 1, no change means 0. What bit string was sent?

WORK

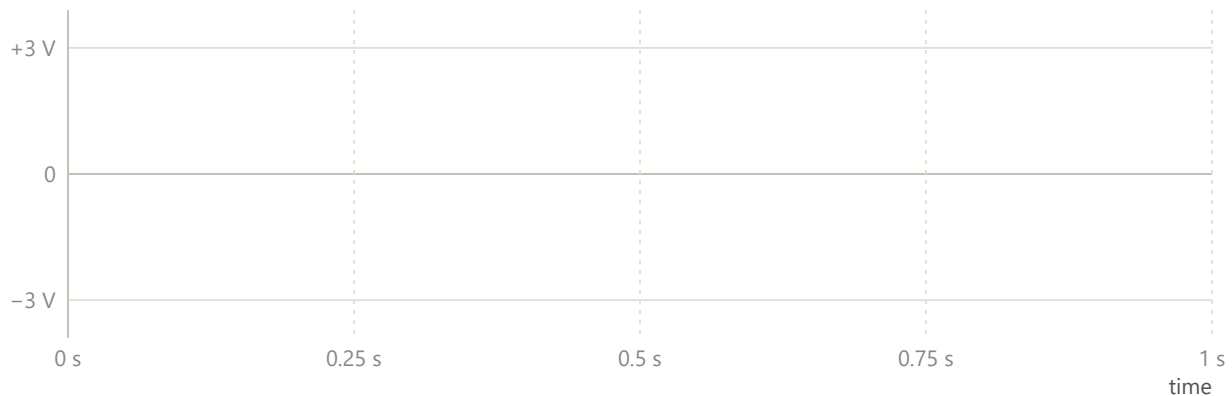
Bits sent = _____

Part D — Draw the waves

Draw on the axes; check the shape bit by bit (or sample by sample) against the key.

12. Draw one second of a sine wave with peak amplitude **3 V**, frequency **1 Hz** and phase **180°** .

Sine wave, 0 to 1 s



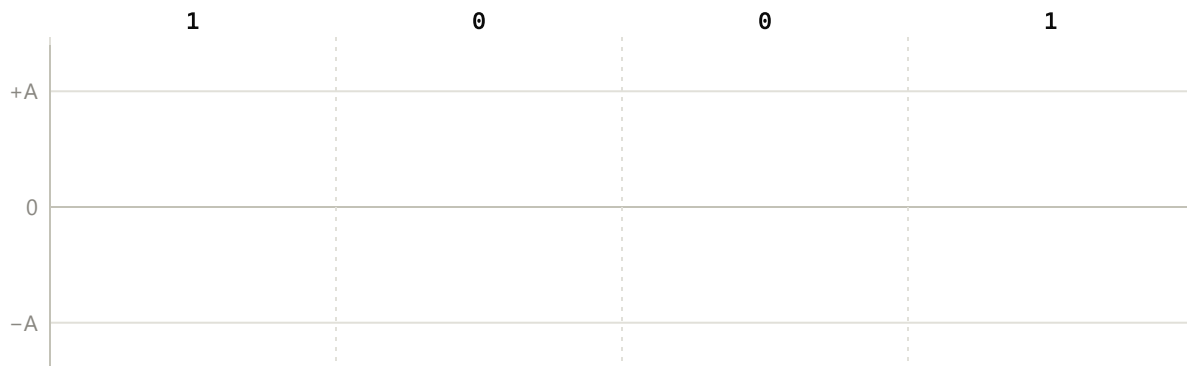
- 13.** Draw the **binary ASK** signal for the bits **10110**: a 1 sends the carrier, a 0 sends no signal (amplitude 0).

Binary ASK for 10110



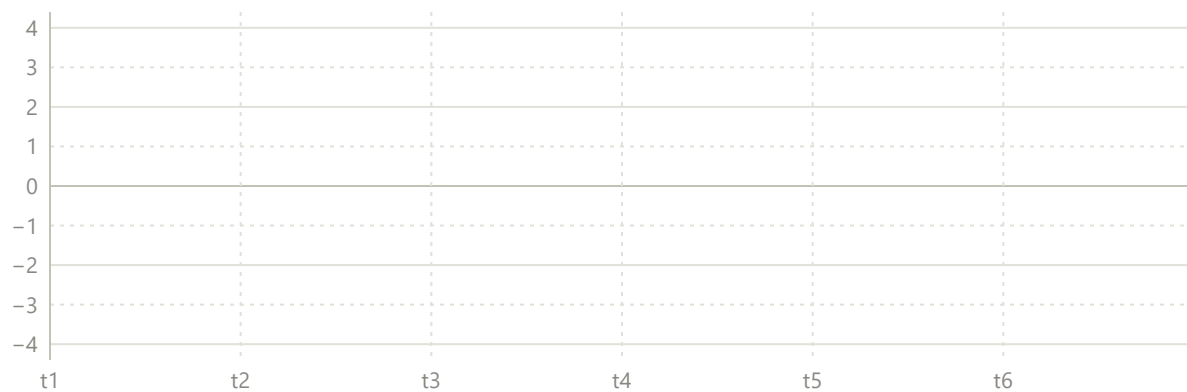
- 14.** Draw the **binary PSK** signal for the bits **1001**: a 1 is the carrier at phase 0° , a 0 is the carrier shifted 180° (draw about 3 cycles per bit).

Binary PSK for 1001



- 15.** A signal between **-4 V** and **4 V** is quantized into **L = 4** zones (zone height $\Delta = 2$ V). Its samples at $t_1 \dots t_6$ are **2.7, -0.4, -3.1, 1.6, 3.5, -1.9 V**. Draw the quantized signal: each sample moved to its zone midpoint and held until the next sample.

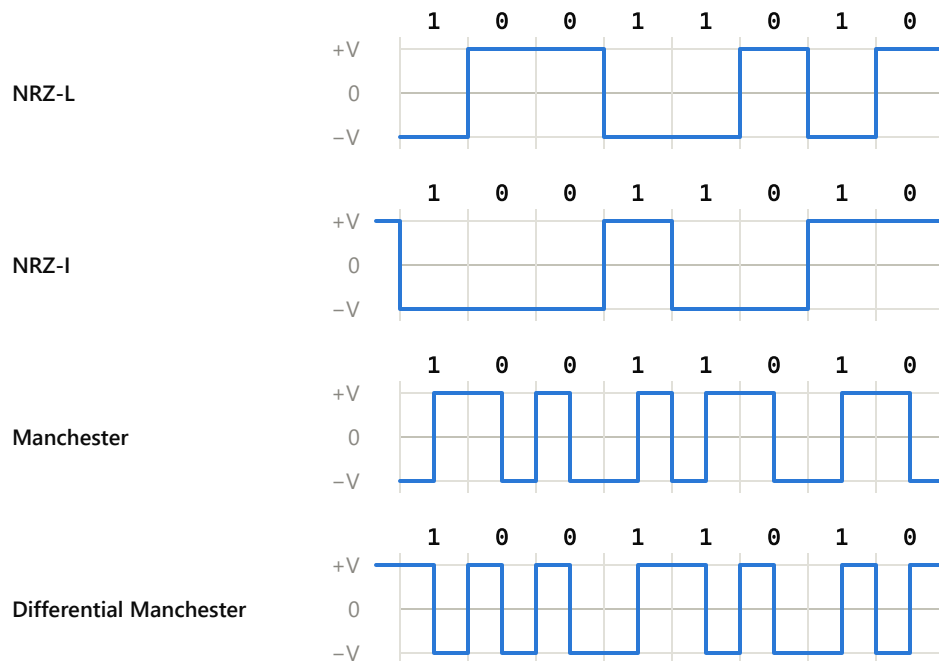
Quantized signal



Answer key

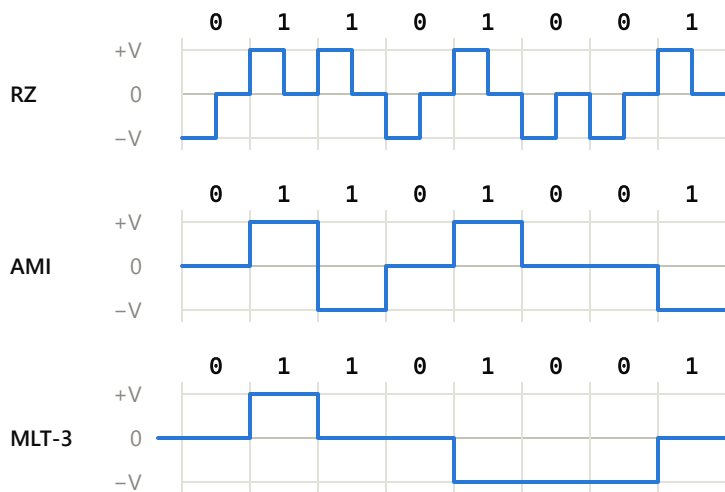
1. PROBLEM

- NRZ-L** — 0 = +V, 1 = -V. L03 p23
- Bit by bit: 1→-, 0→+, 0→+, 1→-, 1→-, 0→+, 1→-, 0→+ — the waveform is - + + - - + - +.
- NRZ-I** — a 1 inverts the level at the start of the bit, a 0 does not; assume the level before the first bit is +V. L03 p23
- Bit by bit: 1→-, 0→-, 0→-, 1→+, 1→-, 0→-, 1→+, 0→+ — the waveform is - - - + - - + +.
- Manchester** — 0 = high→low, 1 = low→high (a transition in the middle of every bit). L03 p29
- Bit by bit: 1→-+, 0→+-, 0→+-, 1→-+, 1→-+, 0→+-, 1→-+, 0→+- — the waveform is (-+)(+-)(+-)(-+)(-+)(+-)(-+)(+-).
- Differential Manchester** — always a transition in the middle of the bit; a 0 also has a transition at the start of the bit, a 1 does not; assume the level before the first bit is +V. L03 p29
- Bit by bit: 1→+-, 0→+-, 0→+-, 1→-+, 1→-+, 0→+-, 1→-+, 0→-+ — the waveform is (+-)(+-)(+-)(-+)(-+)(+-)(-+)(-+).



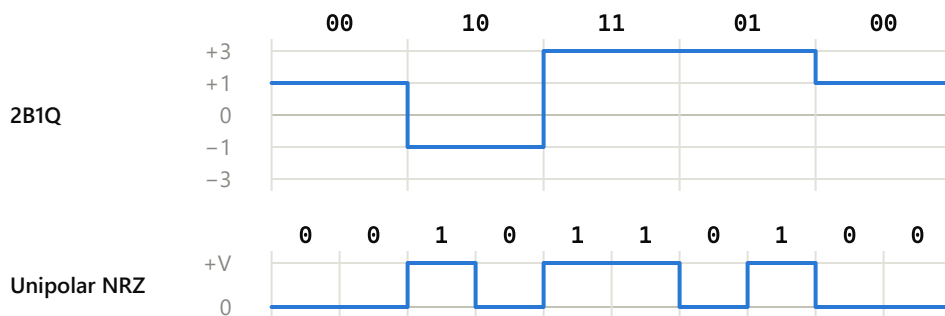
2. PROBLEM

- RZ** — a 1 is +V then 0 V, a 0 is -V then 0 V (every bit returns to zero in the middle). L03 p27
- Bit by bit: 0→-0, 1→+0, 1→+0, 0→-0, 1→+0, 0→-0, 0→-0, 1→+0 — the waveform is (-0)(+0)(+0)(-0)(+0)(-0)(-0)(+0).
- AMI** — 0 = 0 V; the 1s alternate +V, -V, ..., starting with +V. L03 p32
- Bit by bit: 0→0, 1→+, 1→-, 0→0, 1→+, 0→0, 0→0, 1→- — the waveform is 0 + - 0 + 0 0 -.
- MLT-3** — a 0 keeps the level; a 1 moves round the cycle 0 → +V → 0 → -V → ...; start at 0 V with the last nonzero level -V, so the first 1 goes to +V. L03 p44
- Bit by bit: 0→0, 1→+, 1→0, 0→0, 1→-, 0→-, 0→-, 1→0 — the waveform is 0 + 0 0 - - - 0.



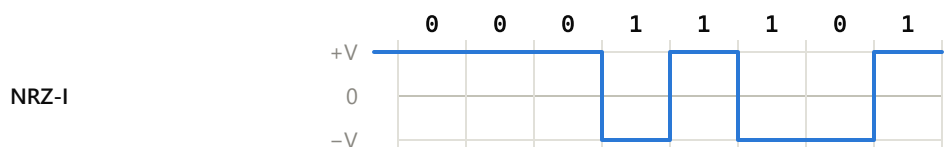
3. PROBLEM

- 2B1Q** — one level per pair of bits from the transition table (previous level positive: 00 → +1, 01 → +3, 10 → -1, 11 → -3; previous level negative: the signs reversed); assume the level before the first pair is positive. L03 p38
- Bit by bit: 00→+1, 10→-1, 11→+3, 01→+3, 00→+1 — the waveform is +1 -1 +3 +3 +1.
- Unipolar NRZ** — 1 = +V, 0 = 0 V. L03 p21
- Bit by bit: 0→0, 0→0, 1→+, 0→0, 1→+, 1→+, 0→0, 1→+, 0→0, 0→0 — the waveform is 0 0 + 0 + + 0 + 0 0.

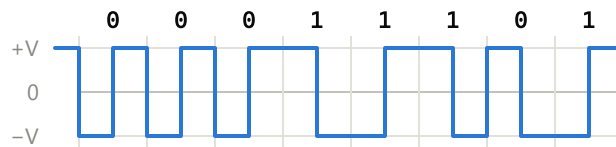


4. PROBLEM

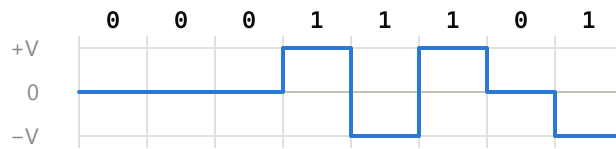
- NRZ-I** — a 1 inverts the level at the start of the bit, a 0 does not; assume the level before the first bit is +V. L03 p23
- Bit by bit: 0→+, 0→+, 0→+, 1→-, 1→+, 1→-, 0→-, 1→+ — the waveform is + + + - + - - +.
- Differential Manchester** — always a transition in the middle of the bit; a 0 also has a transition at the start of the bit, a 1 does not; assume the level before the first bit is +V. L03 p29
- Bit by bit: 0→-+, 0→-+, 0→-+, 1→+-, 1→-+, 1→-+, 0→+-, 1→-+ — the waveform is (-+)(-+)(-+)(+-)(+-)(+-)(+ -)(-+).
- AMI** — 0 = 0 V; the 1s alternate +V, -V, ..., starting with +V. L03 p32
- Bit by bit: 0→0, 0→0, 0→0, 1→+, 1→-, 1→+, 0→0, 1→- — the waveform is 0 0 0 + - + 0 -.



Differential Manchester



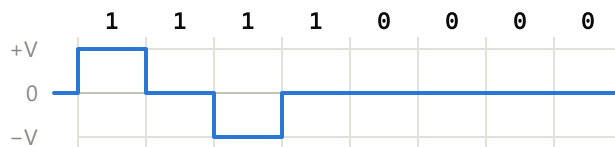
AMI



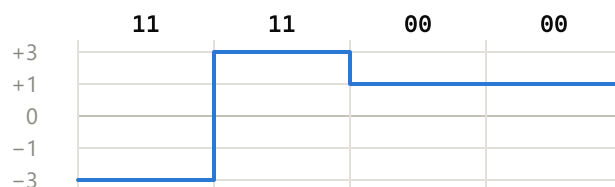
5. PROBLEM

- MLT-3** — a 0 keeps the level; a 1 moves round the cycle $0 \rightarrow +V \rightarrow 0 \rightarrow -V \rightarrow \dots$; start at 0 V with the last nonzero level $-V$, so the first 1 goes to $+V$. [L03 p44](#)
- Bit by bit: $1 \rightarrow +$, $1 \rightarrow 0$, $1 \rightarrow -$, $1 \rightarrow 0$, $0 \rightarrow 0$, $0 \rightarrow 0$, $0 \rightarrow 0$, $0 \rightarrow 0$ — the waveform is $+ \ 0 \ - \ 0 \ 0 \ 0 \ 0$.
- 2B1Q** — one level per pair of bits from the transition table (previous level positive: $00 \rightarrow +1$, $01 \rightarrow +3$, $10 \rightarrow -1$, $11 \rightarrow -3$; previous level negative: the signs reversed); assume the level before the first pair is positive. [L03 p38](#)
- Bit by bit: $11 \rightarrow -3$, $11 \rightarrow +3$, $00 \rightarrow +1$, $00 \rightarrow -1$ — the waveform is $-3 \ +3 \ +1 \ -1$.
- Manchester** — $0 = \text{high} \rightarrow \text{low}$, $1 = \text{low} \rightarrow \text{high}$ (a transition in the middle of every bit). [L03 p29](#)
- Bit by bit: $1 \rightarrow +$, $1 \rightarrow +$, $1 \rightarrow +$, $1 \rightarrow +$, $0 \rightarrow -$, $0 \rightarrow -$, $0 \rightarrow -$, $0 \rightarrow -$ — the waveform is $(-+)(-+)(-+)(-+)(+-)(+-)(+-)(+-)$.

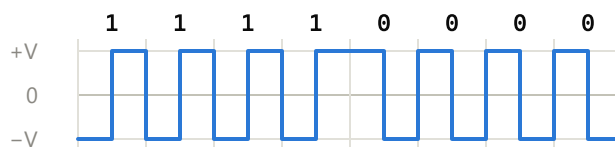
MLT-3



2B1Q

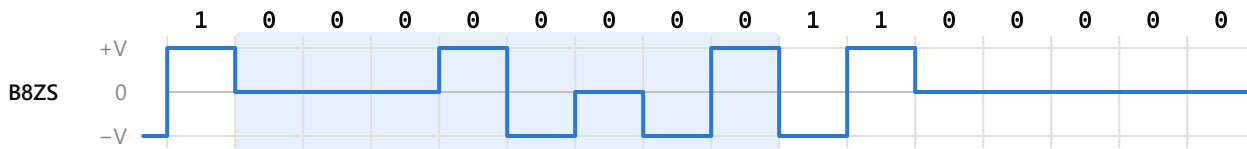


Manchester



6. PROBLEM

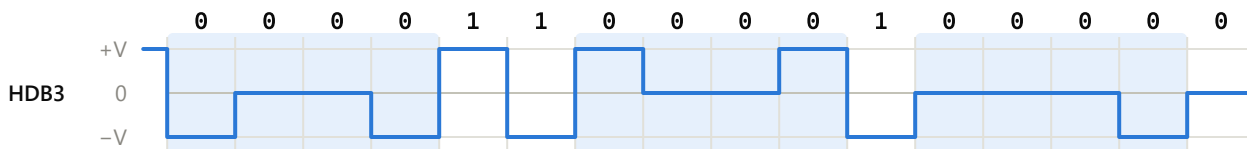
- Scrambling works on top of AMI: $0 = 0 \text{ V}$ and the 1s alternate in polarity. The last pulse before the stream was negative ($-V$), so the next 1 is $+$. [L03 p58](#)
- B8ZS replaces eight consecutive 0s with **000VB0VB**: V = a violation (the same polarity as the previous pulse), B = a normal bipolar pulse (the opposite polarity). [L03 p59](#)
- Bits 2–9 are eight 0s and the pulse before them is positive ($+V$): $V = +$, $B = -$, $0 = -$ (the same as the B before it), $B = + \rightarrow 0 \ 0 \ 0 \ + \ - \ 0 \ - \ +$. [L03 p59](#)
- The transmitted pulses are $+ \ 0 \ 0 \ 0 \ + \ - \ 0 \ - \ + \ - \ + \ 0 \ 0 \ 0 \ 0 \ 0$ — no run of eight zeros is left.



Pulse before the stream: -. Substitutions: bits 2–9 → 000VB0VB. Marks: V at bit 5, B at bit 6, V at bit 8, B at bit 9.

7. PROBLEM

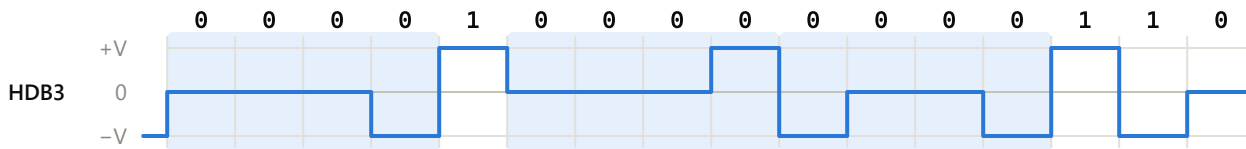
- Scrambling works on top of AMI: 0 = 0 V and the 1s alternate in polarity. The last pulse before the stream was positive (+V), so the next 1 is -. L03 p58
- HDB3 replaces four consecutive 0s: count the nonzero pulses since the last substitution — **even** → **B00V**, **odd** → **000V** (the total becomes even either way). L03 p60
- Substitution 1 (bits 1–4): 0 nonzero pulses since the start → even → **B00V** = - 0 0 - (B alternates from the previous pulse, V repeats B). L03 p61
- Substitution 2 (bits 7–10): 2 nonzero pulses since the last substitution → even → **B00V** = + 0 0 + (B alternates from the previous pulse, V repeats B). L03 p61
- Substitution 3 (bits 12–15): 1 nonzero pulse since the last substitution → odd → **000V** = 0 0 0 - (V repeats the previous pulse). L03 p61
- The transmitted pulses are - 0 0 - + - + 0 0 + - 0 0 0 - 0 — no run of four zeros is left.



Pulse before the stream: +. Substitutions: bits 1–4 → B00V (even pulses since last substitution); bits 7–10 → B00V (even pulses since last substitution); bits 12–15 → 000V (odd pulses since last substitution). Marks: B at bit 1, V at bit 4, B at bit 7, V at bit 10, V at bit 15.

8. PROBLEM

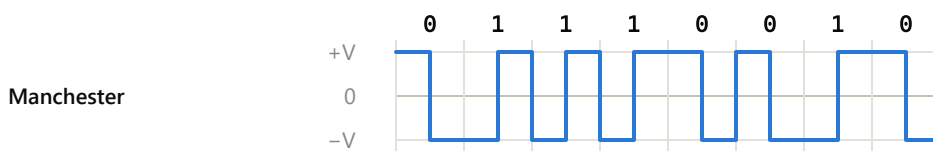
- Scrambling works on top of AMI: 0 = 0 V and the 1s alternate in polarity. The last pulse before the stream was negative (-V), so the next 1 is +. L03 p58
- HDB3 replaces four consecutive 0s: count the nonzero pulses since the last substitution — **even** → **B00V**, **odd** → **000V** (the total becomes even either way). L03 p60
- The problem says the count was already odd at the start of the stream, so counting starts at 1.
- Substitution 1 (bits 1–4): 1 nonzero pulse since the start → odd → **000V** = 0 0 0 - (V repeats the previous pulse). L03 p61
- Substitution 2 (bits 6–9): 1 nonzero pulse since the last substitution → odd → **000V** = 0 0 0 + (V repeats the previous pulse). L03 p61
- Substitution 3 (bits 10–13): 0 nonzero pulses since the last substitution → even → **B00V** = - 0 0 - (B alternates from the previous pulse, V repeats B). L03 p61
- The transmitted pulses are 0 0 0 - + 0 0 0 + - 0 0 - + - 0 — no run of four zeros is left.



Pulse before the stream: -. Substitutions: bits 1–4 → 000V (odd pulses since last substitution); bits 6–9 → 000V (odd pulses since last substitution); bits 10–13 → B00V (even pulses since last substitution). Marks: V at bit 4, V at bit 9, B at bit 10, V at bit 13.

9. PROBLEM

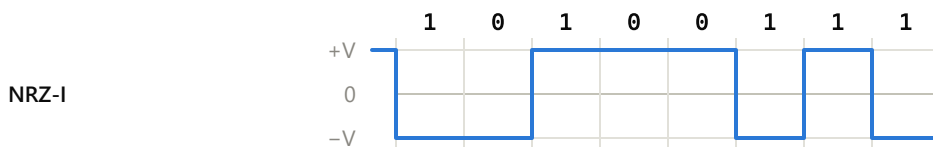
- Manchester: 0 = high→low, 1 = low→high (a transition in the middle of every bit). L03 p29
- Read the signal one bit at a time: (+-) → 0; (-+) → 1; (-+) → 1; (-+) → 1; (+-) → 0; (+-) → 0; (-+) → 1; (+-) → 0.
- The data bits are **01110010**.



Bits sent = **01110010**

10. PROBLEM

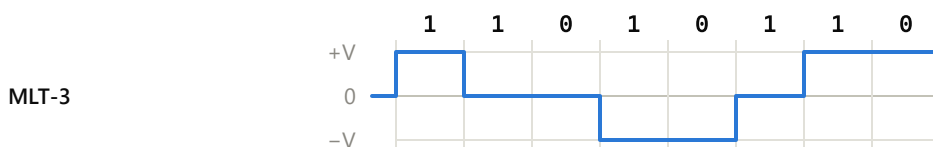
- NRZ-I: a 1 inverts the level at the start of the bit, a 0 does not; assume the level before the first bit is +V. L03 p23
- Read the signal one bit at a time: - (changed) → 1; - (same) → 0; + (changed) → 1; + (same) → 0; + (same) → 0; - (changed) → 1; + (changed) → 1; - (changed) → 1.
- The data bits are **10100111**.



Bits sent = **10100111**

11. PROBLEM

- MLT-3: a 0 keeps the level; a 1 moves round the cycle 0 → +V → 0 → -V → ...; start at 0 V with the last nonzero level -V, so the first 1 goes to +V. L03 p44
- Read the signal one bit at a time: + (changed) → 1; 0 (changed) → 1; 0 (same) → 0; - (changed) → 1; - (same) → 0; 0 (changed) → 1; + (changed) → 1; + (same) → 0.
- The data bits are **11010110**.

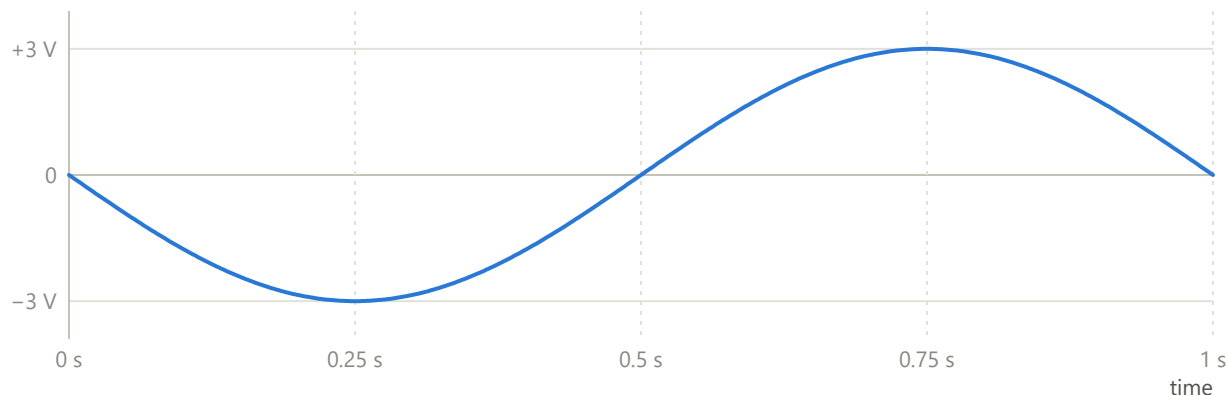


Bits sent = **11010110**

12. PROBLEM

1. Period: $T = 1/f = 1/1 \text{ Hz} = 1 \text{ s}$, so 1 cycle fits in 1 s. L02 p7
2. Phase 180° : the wave starts **at 0, going down**. L02 p8
3. Peak amplitude 3 V: the crests reach +3 V and the troughs -3 V. L02 p7

Sine wave, 0 to 1 s

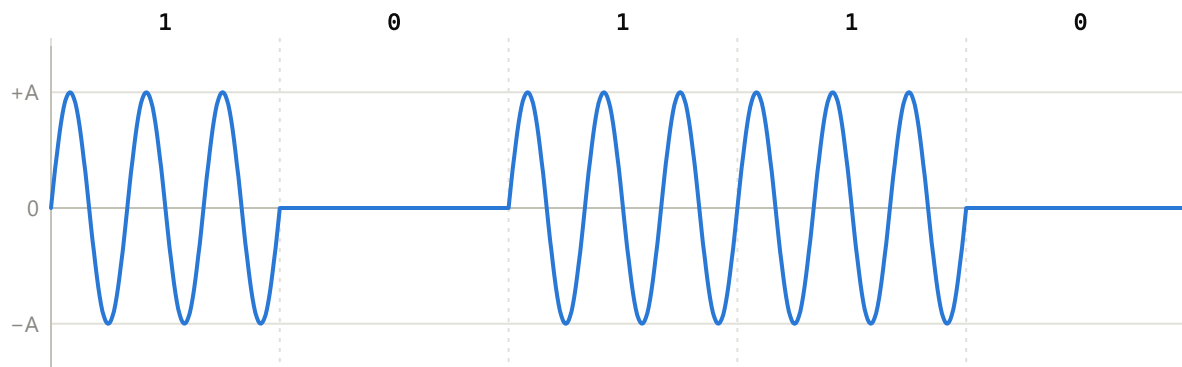


- At $t = 0$ the wave starts at 0, going down (phase 180°).
- 1 full cycle in 1 s — one cycle every $T = 1/f = 1 \text{ s}$.
- Crests at +3 V and troughs at -3 V.

13. PROBLEM

1. The rule for binary ASK: a 1 sends the carrier, a 0 sends no signal (amplitude 0). L04 p8
2. Bit by bit: 1 → carrier, 0 → nothing, 1 → carrier, 1 → carrier, 0 → nothing.
3. A binary scheme carries one bit per signal element ($r = 1$), so every bit cell has the same width.

Binary ASK for 10110



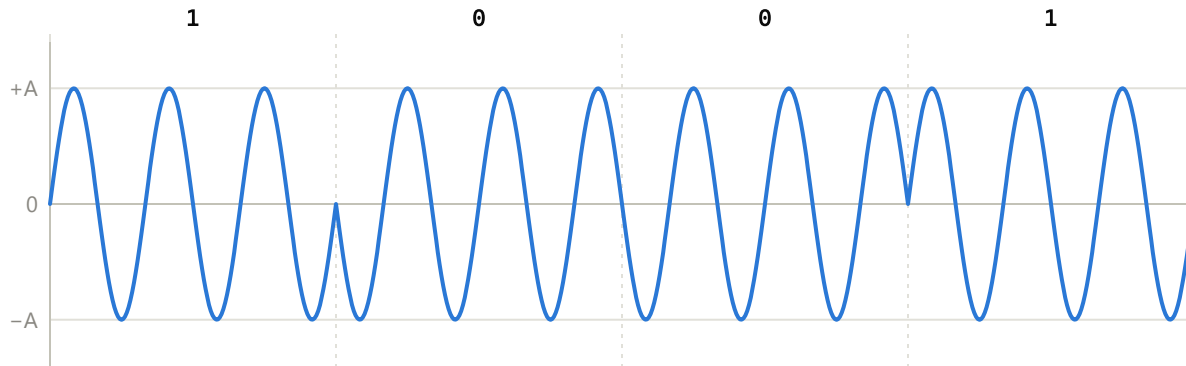
- Every 1-bit is the carrier at full amplitude.
- Every 0-bit is a flat line at 0 (no signal).
- The same number of carrier cycles in every 1-bit.

14. PROBLEM

1. The rule for binary PSK: a 1 is the carrier at phase 0° , a 0 is the carrier shifted 180° (draw about 3 cycles per bit). L04 p22
2. Bit by bit: 1 → 0° , 0 → 180° , 0 → 180° , 1 → 0° .

3. A binary scheme carries one bit per signal element ($r = 1$), so every bit cell has the same width.

Binary PSK for 1001

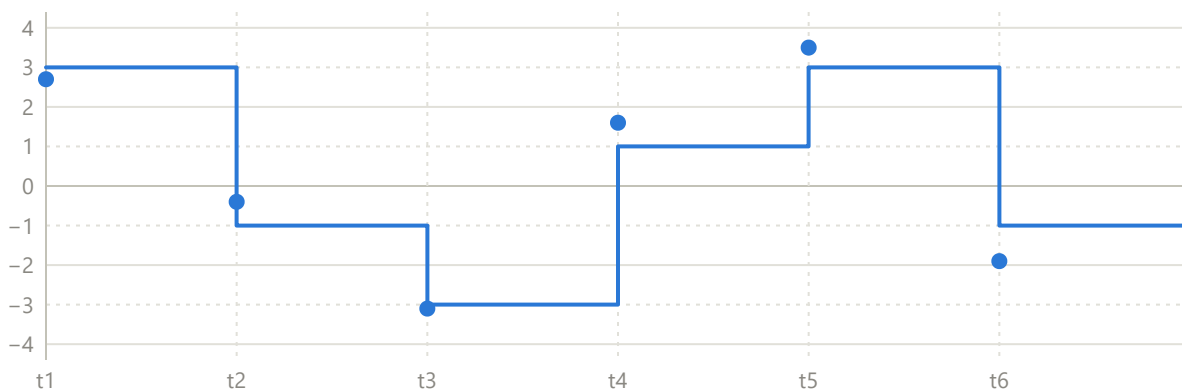


- Every 1-bit starts at 0 going up (phase 0°).
- Every 0-bit starts at 0 going down (phase 180°).
- The phase flips at each 0 $\leftrightarrow$ 1 boundary; amplitude and frequency stay the same.

15. PROBLEM

1. Zone height $\Delta = (\max - \min)/L = (4 - (-4))/4 = 2 \text{ V}$; the zone midpoints are $-3, -1, 1, 3 \text{ V}$. L03 p72
2. Each sample goes to the midpoint of its zone: $2.7 \rightarrow 3, -0.4 \rightarrow -1, -3.1 \rightarrow -3, 1.6 \rightarrow 1, 3.5 \rightarrow 3, -1.9 \rightarrow -1$. L03 p76
3. Holding each value until the next sample arrives (as the decoder's hold circuit does) turns the samples into a staircase. L03 p81

Quantized signal



- Each sample moved to its zone midpoint: $2.7 \rightarrow 3, -0.4 \rightarrow -1, -3.1 \rightarrow -3, 1.6 \rightarrow 1, 3.5 \rightarrow 3, -1.9 \rightarrow -1 \text{ V}$.
- Flat steps: each value held for one sample interval (a staircase that changes only at the sampling instants).

END OF DOCUMENT: drills · build r1